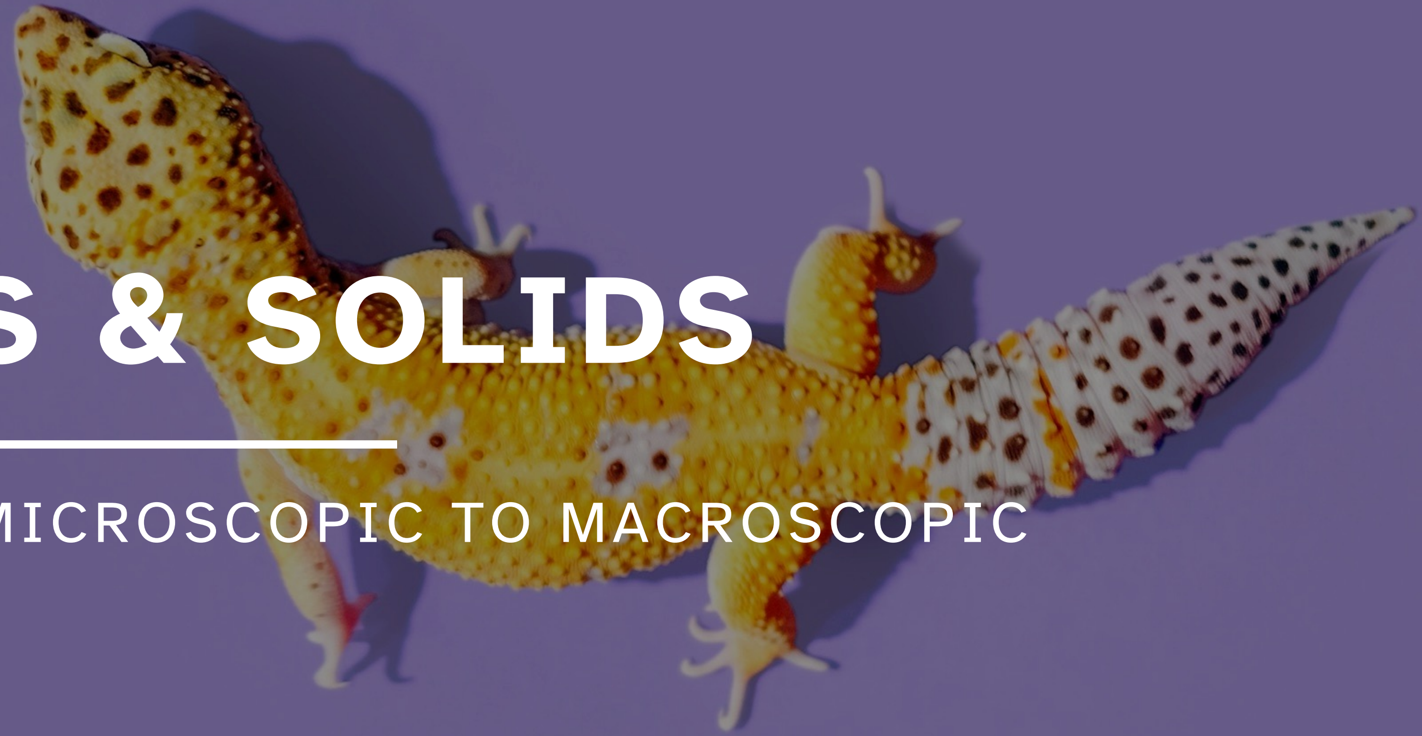
12A
LIQUIDS & SOLIDS



UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

MIOY T. HUYNH

LEARNING OBJECTIVES

CHAPTER 12

1. List and describe the four types of intermolecular forces and identify the predominant intermolecular force in a substance.
2. Define and apply terms related to phase changes.
3. Calculate energy changes related to phase changes.
4. Interpret heating curves.
5. Predict physical properties (relative vapor pressures and relative phase change temperatures) based on strengths of intermolecular forces.
6. Define, describe, and interpret phase diagrams.
7. Describe the structures of various cubic unit cells.

Where did we start and where are we going?

Chemistry in Context

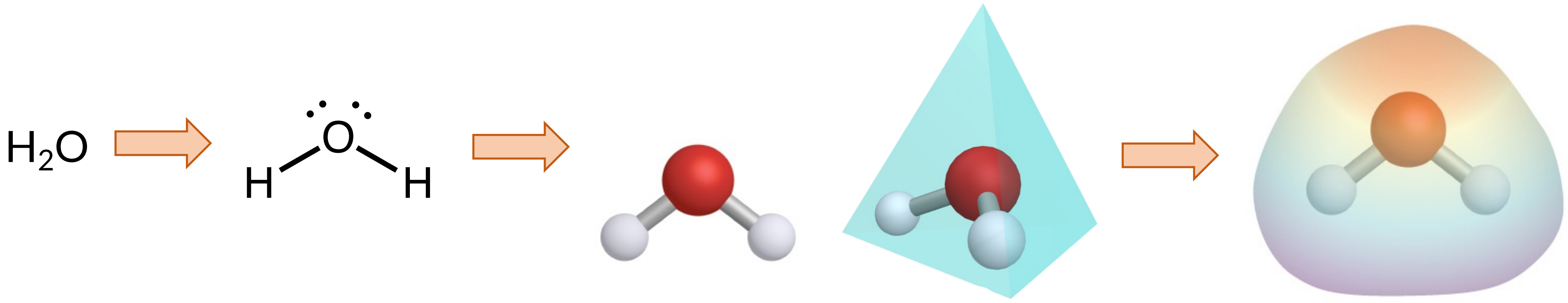
What is our overall goal?

Chemical Formulas → Atomic Structure → Chemical Bonding → Molecular Structure → Molecular Properties

Intermolecular Forces

12.1

Everything discussed so far has been regarding **intra**molecular forces, which are the chemical ***bonds*** that hold ***atoms*** together in a ***single molecule***.



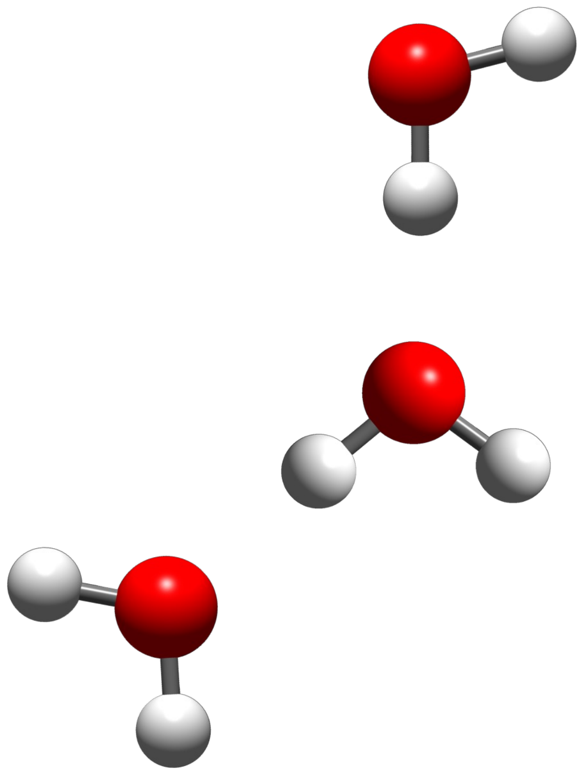
To think about the properties of *macroscopic or bulk samples*, we need to consider more than one molecule at a time.

Intermolecular Forces

12.1

Intermolecular forces (IMFs) are the attractive forces that hold *individual molecules* together in a *sample*.

- IMFS are ***MUCH*** weaker than chemical bonds.
- Arise from *electrostatic* interactions between "charges" on adjacent molecules.



It is critical you differentiate *intra*molecular and *inter*molecular forces at a *microscopic* level!

Intermolecular Forces

12.1

Types of IMFS depend mostly on molecular polarity!

	LDF	DD	HB	ID
	London Dispersion Forces	**Dipole-Dipole Attractions**	**Hydrogen Bonding Interactions**	**Ion-Dipole Attractions**
Nonpolar molecules & noble gases				
Polar molecules				
Polar molecules that have H—O, H—N, or H—F bonds				
Mixture of polar molecules + soluble ionic compounds				

Intermolecular Forces

12.1

London dispersion forces LDF

- Present among all atoms and molecules.
- Generally, the weakest IMF.
- Strongest IMF when considering samples of *nonpolar molecules* or *noble gases*.
- Due to *temporary, instantaneous* dipoles: uneven e^- distribution

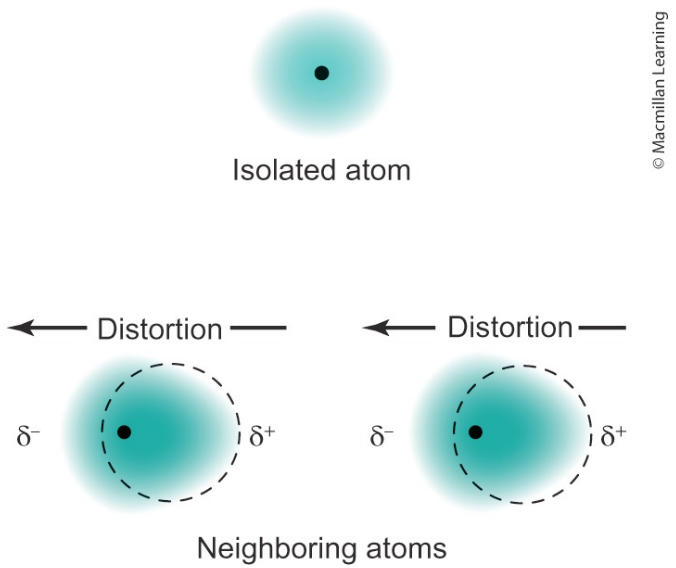


FIGURE 12.8

Intermolecular Forces

12.1

Strength of London dispersion forces depends on a molecule's polarizability.

- Ease with which the electrons (e^-) in an atom or molecule can be *distorted*.
- More polarizable → frequent and stronger temporary dipoles → *stronger* LDFs

What makes a molecule *more polarizable??*

- Larger mass →
- Larger atoms/molecules →
- Longer, more "oblong" molecules →

Intermolecular Forces

12.1

Dipole-dipole attractions DD

- Polar molecules have *permanent* molecular dipoles.
- The δ+ of one molecular dipole is attracted to the permanent δ– of the molecular dipole on another molecule.
- *Common examples*: HCl, CH_2Cl_2, any pure sample of a polar molecule

FIGURE 12.3

Intermolecular Forces

12.1

Hydrogen-bonding interactions HB

- Special and *stronger* form of dipole–dipole attractions
- Strict Requirements:
  - HB *donor* molecule: polar molecule with a H atom *bonded* to N, O, or F
  - HB *acceptor* molecule: polar molecule with a lone pair on N, O, or F

H is very small and N, O, F are very small and electronegative (EN)

Very polar bonds (H—N, H—O, H—F have large ΔEN) → very strong molecular dipoles

Intermolecular Forces

12.1

Hydrogen-bonding interactions require us to think about the locations and orientations of lone pair electrons.

(a) H_2O molecules

(b) NH_3 molecules

(c) HF molecules

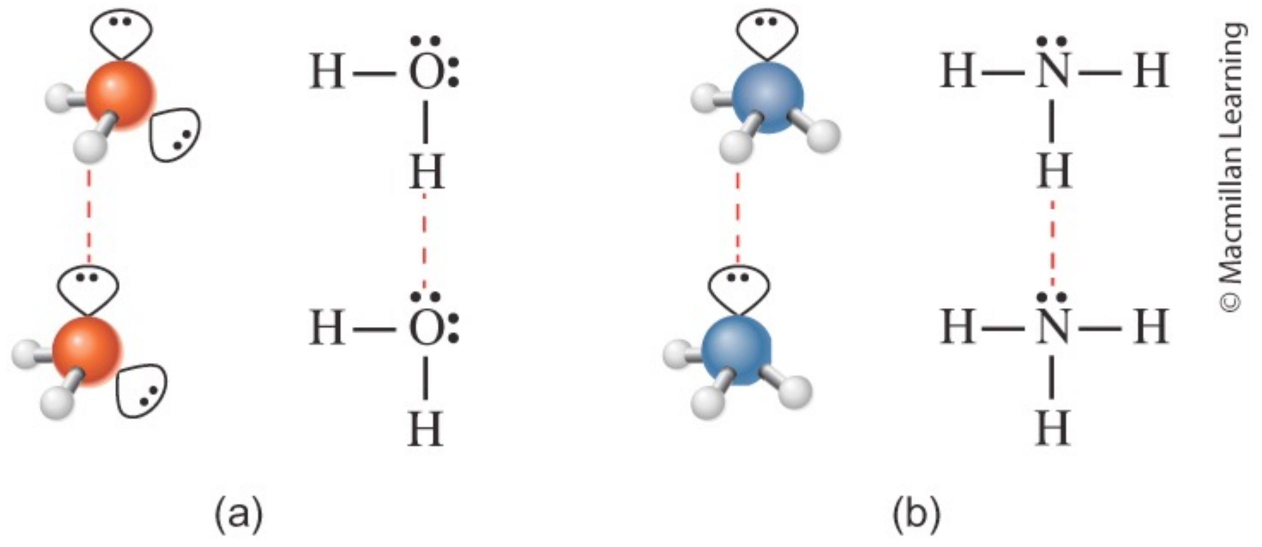


FIGURE 12.5

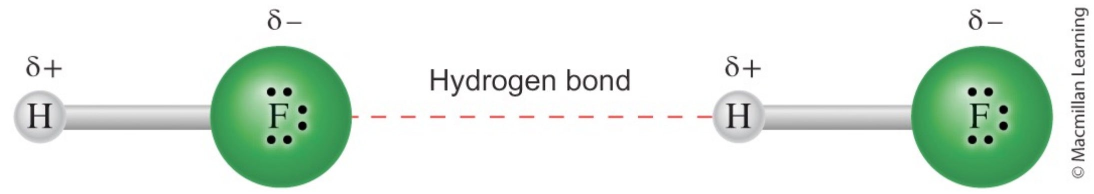


FIGURE 12.4

Intermolecular Forces

12.1

Ion-dipole attractions ID

- Typically, the strongest of the IMFs.
- An ion (cation or anion) is attracted to the partial charge (of the opposite sign) of the permanent dipole in a polar molecule.
- *Most common examples*: soluble ionic compounds dissolved in water

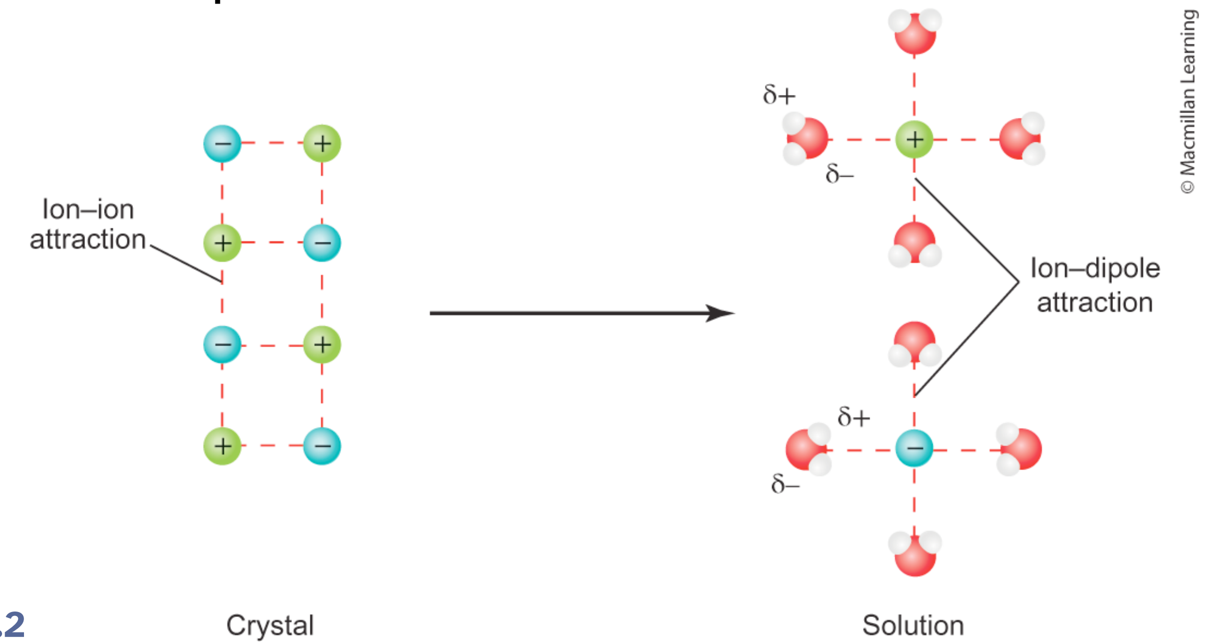


FIGURE 12.2

EXAMPLE PROBLEM

Learning Objective(s): 12.1

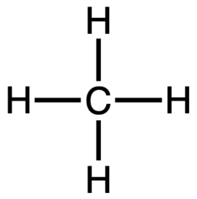
Can molecules of CH_3OCH_3 hydrogen bond to molecules of HF?

Can two molecules of CH_3OCH_3 hydrogen bond to one another?

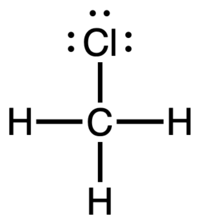
IN-CLASS QUESTION 1

Learning Objective(s): 12.1

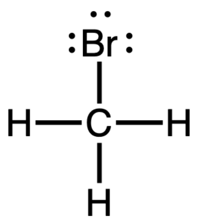
Identify the primary/dominant intermolecular force in pure samples of each of the following substances.



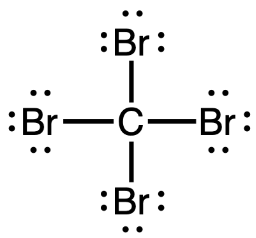
Substance A
CH_4



Substance B
CH_3Cl



Substance C
CH_3Br



Substance D
CBr_4

EXAMPLE PROBLEM

Learning Objective(s): 12.1

Determine the primary/dominant intermolecular force that is present in pure samples of each of the following compounds.

- CH_3COOH
- H_2S
- SiH_4
- CH_2F_2

Intermolecular Forces

12.1

Physical molecular properties can be predicted from the types and strengths of IMFs.

Only IMFs are broken during phase changes (no chemical bonds are broken).

Intermolecular Forces

12.1

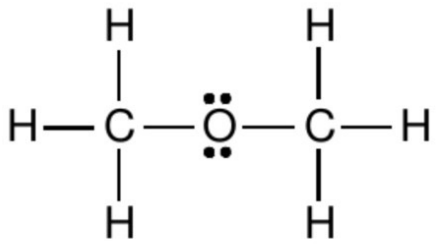
Relating polarizability, strength of LDFs, and melting/boiling points for the halogens.

Molecule	Total Electrons	Molar Mass	Atomic Radius	Boiling Point	Melting Point
F_2		~38 g/mol	60 pm	85 K	53 K
Cl_2		~71 g/mol	100 pm	239 K	172 K
Br_2		~160 g/mol	117 pm	332 K	266 K
I_2		~254 g/mol	136 pm	457 K	387 K

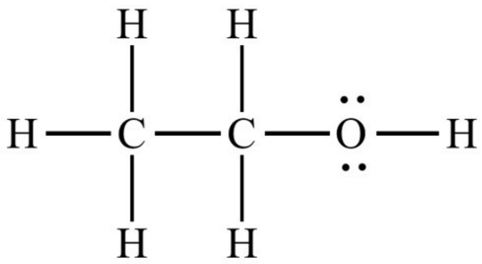
BONUS QUESTION

Assessed on: Fall 2021, Exam 3

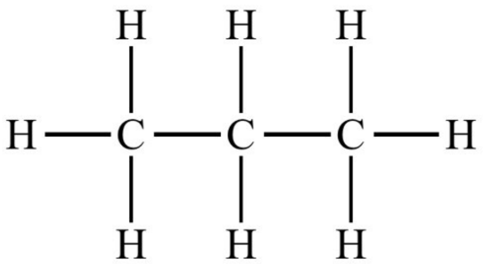
Predict and arrange the boiling points of the following samples from lowest to highest.



CH_3OCH_3



CH_3CH_2OH



$CH_3CH_2CH_3$